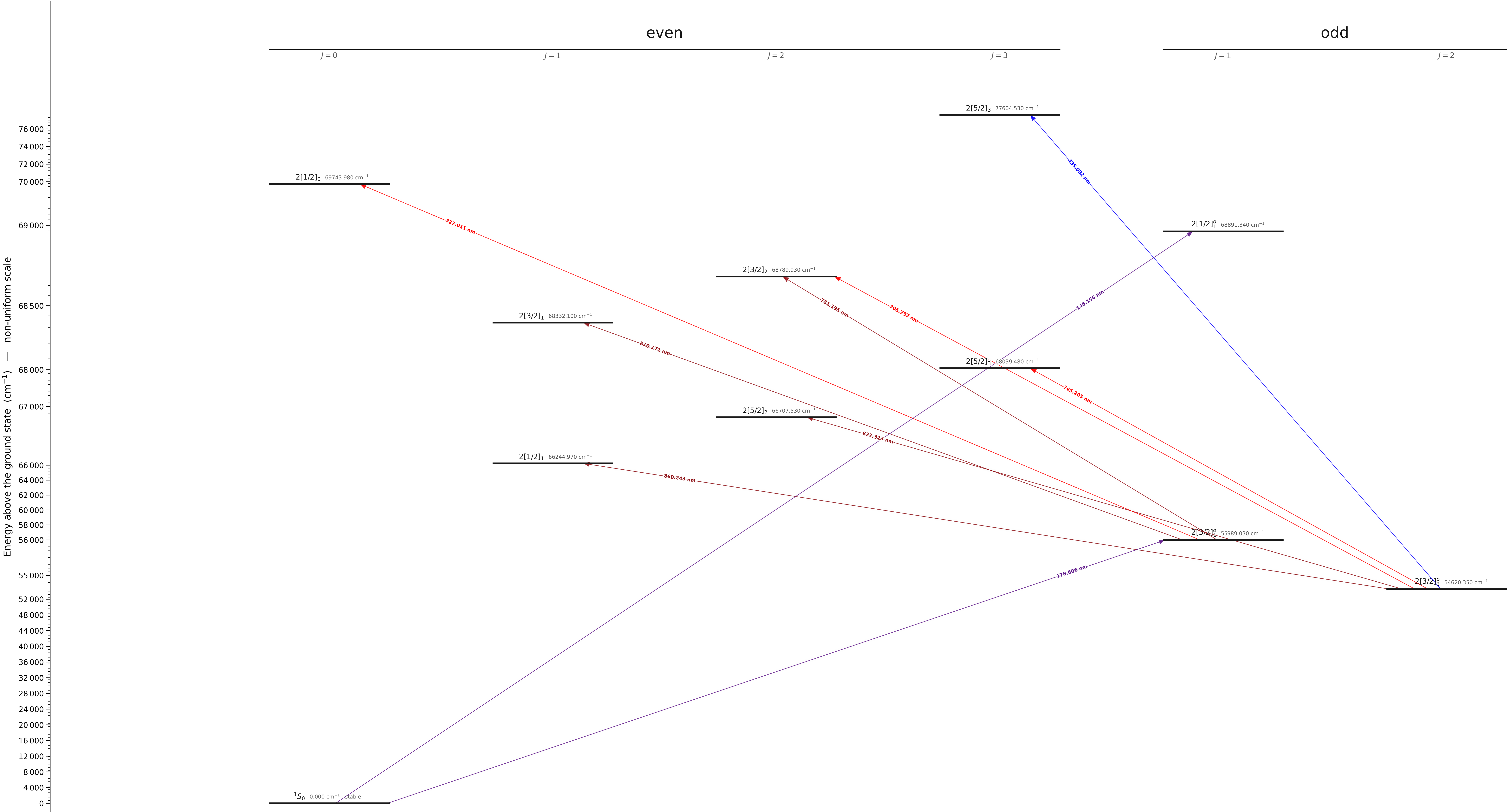
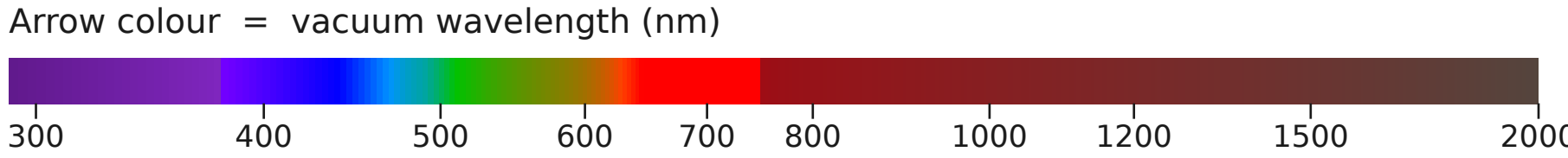


Radon

Rn (Rn I). Strongest classified transitions below 2 μm (NIST): 11 levels, 10 lines.

Rn⁺ ionisation limit 86692.50 cm⁻¹ = 10.74850 eV (115.350 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J'\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2[3/2] ₂ - 2[5/2] ₃	745.2048	744.9996	402.2954	-	-	-	-	the line of the only laser-spectroscopy study of radon: c...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).